

Xunjian Yin

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Duke University, Durham - 27705, US.

RESEARCH INTERESTS

#Recursive Self-Improvement #Continual Learning #Long Horizon Agent

EDUCATION

- **Duke University** 08.2025 - 06.2029 (expected)
Ph.D. in Computer Science, co-advised by Prof. [Shuyan Zhou](#) and Prof. [Bhuwan Dhingra](#)
- **Peking University** 09.2022 - 06.2025
M.S. in Computer Science, supervised by Prof. [Xiaojun Wan](#) GPA: 3.81/4.00
- **Peking University** 09.2018 - 06.2022
B.S. in Computer Science GPA: 3.64/4.00

SELECTED PUBLICATIONS

Preprints:

- [1] **From Atomic to Composite: RL Enables Generalization in Complementary Reasoning** *Arxiv* [\[link\]](#)
Sitao Cheng, **Xunjian Yin**, Ruiwen Zhou, ..., Liangming Pan, William Yang Wang, Victor Zhong.
- [2] **ContraSolver: Self-Alignment of LLMs by Resolving Internal Preference Contradictions** *Arxiv* [\[link\]](#)
Xu Zhang*, **Xunjian Yin***, Xiaojun Wan.

Conference paper:

- [1] **Constructing Challenging Browser-Use Tasks by Controlled Environment Interventions**
NeurIPS 2026 [\[link\]](#)
Xunjian Yin*, Tianchen Guan*, Jinao Wang, ..., Bhuwan Dhingra, Xiangjun Wang, Shuyan Zhou.
- [2] **Gödel Agent: A Self-Referential Agents Framework for Recursively Self-Improvement**
ACL 2025 [\[link\]](#)
Xunjian Yin, Xinyi Wang, Liangming Pan, Xiaojun Wan, William Wang.
- [3] **LEDOM: Reverse Language Model**
ACL 2026 [\[link\]](#)
Xunjian Yin, Sitao Cheng, Yuxi Xie, ..., Xinyi Wang, Liangming Pan, William Wang, Xiaojun Wan.
- [4] **The Geometry of Reasoning: Flowing Logics in Representation Space**
ICLR 2026 [\[link\]](#)
Yufa Zhou*, Yixiao Wang*, **Xunjian Yin***, Shuyan Zhou, Anru R. Zhang.
- [5] **DAMON: A Dialogue-Aware MCTS Framework for Jailbreaking Large Language Models**
EMNLP 2025 [\[link\]](#)
Xu Zhang, **Xunjian Yin**, Dinghao Jing, Huixuan Zhang, Xinyu Hu, Xiaojun Wan.
- [6] **ChemAgent: Self-updating Memories in LLMs Improves Chemical Reasoning**
ICLR 2025 [\[link\]](#)
Xiangru Tang, Tianyu Hu, MUYANG YE, Yanjun Shao, **Xunjian Yin**, ..., Arman Cohan, Mark Gerstein
- [7] **Benchmarking Knowledge Boundary for LLMs: A Different Perspective on Model Evaluation**
ACL 2024 [\[link\]](#)
Xunjian Yin*, Xu Zhang*, Jie Ruan, Xiaojun Wan.
- [8] **History Matters: Temporal Knowledge Editing in Large Language Models**
AAAI 2024 [\[link\]](#)
Xunjian Yin, Jin Jiang, Liming Yang, Xiaojun Wan.
- [9] **ALCUNA: Large Language Models Meet New Knowledge**
EMNLP 2023 [\[link\]](#)
Xunjian Yin*, Baizhou Huang*, Xiaojun Wan.
- [10] **How Do Seq2Seq Models Perform on End-to-End Data-to-Text Generation?**
ACL 2022 [\[link\]](#)
Xunjian Yin, Xiaojun Wan.
- [11] **Themis: A Reference-free NLG Evaluation Model with Flexibility and Interpretability**
EMNLP 2024 [\[link\]](#)
Xinyu Hu, Li Lin, Mingqi Gao, **Xunjian Yin**, Xiaojun Wan.

RESEARCH EXPERIENCE

- **Google DeepMind** 05.2026 - 09.2026
Advisor: Dr. Jingmeng Rao Role: Student Researcher Mountain View, USA
 - **Project:** Working Memory Management for Long Horizon Agentic Tasks
 - * We designed a method that enables internal agents to autonomously and dynamically manage their context and retrieve necessary information from their history without loss.
- **Amazon AGI SF Lab** 10.2025 - 05.2026
Manager: Xiangjun Wang Role: Research Collaborator San Francisco, USA
 - **Project:** World Models and Benchmarks for Computer-Use Agents
 - * We build world models for computer-use agents and benchmarks to evaluate them.
- **University of California, Santa Barbara (NLP Group)** 06.2024 - 02.2025
Advisor: Prof. William Wang Role: Visiting Research Scholar California, USA
 - **Project 1:** Reverse Language Model (RLM) Pre-training, Evaluation, Analysis and Applications
 - * We trained RLMs (2B-7B) from scratch, using a 500 billion tokens and the last-token prediction approach.
 - **Project 2:** Gödel Agent: A Self-Referential Agents Framework for Recursively Self-Improvement
 - * We developed a self-referential agent, which is capable of reading and modifying its own logic and code.
- **Microsoft Research Asia (NLC Group)** 02.2022 - 08.2022
Advisor: Dr. Shuming Ma and Dr. Kai Chen Role: Research Intern Beijing, China
 - **Project:** Pre-training with Curriculum Learning (CL)
 - * We investigated the application of CL during pretraining, which accelerates the convergence of the pretraining.
- **Wangxuan Institute of Computer Technology, Peking University** 10.2020 - 06.2022
Advisor: Prof. Xiaojun Wan Role: Research Assistant Beijing, China
 - **Project 1:** Analysis of Seq2Seq Models on End-to-End Data-to-Text Generation
 - * We find that the most advanced models do not always yield the best performance and larger models are better.
 - **Project 2:** Enhancing Language Models with k NN for Grammar Error Correction (GEC)
 - * A k NN algorithm with robust information derived from character phonetics and shapes to augment LMs.
- **Institute of Computational Linguistics, Peking University** 04.2020 - 11.2021
Advisor: Prof. Yunfang Wu Role: Research Assistant Beijing, China
 - **Project:** Multi-Task Learning for Grammar Error Correction
- **Institute of Computational Linguistics, Peking University** 07.2019 - 12.2019
Advisor: Prof. Sujian Li Role: Research Assistant Beijing, China
 - **Project:** Building Benchmark for Mathematical Olympiad Problems

HONORS AND AWARDS

- **Gold Reviewer Award, ICML 2026** 05.2026
- **Outstanding Graduate of Beijing, Beijing Municipality** 06.2025
- **Outstanding Graduate of Peking University, Peking University** 06.2025
- **Outstanding Master's Thesis Award of WICT, Peking University** 06.2025
- **Merit Student, Peking University** 09.2024
- **Guotai Junan Scholarship, Peking University** 09.2024
- **Wang Xuan Scholarship, Peking University** 09.2022
- **Award for Research Excellence, Peking University** 09.2021, 09.2022, 09.2023
- **Award for Academic Excellence, Peking University** 09.2019, 09.2020
- **Outstanding Student of Shandong Province, Shandong Province** 04.2018

SERVICE

- **Teaching Assistant:** Duke University (2026); Peking University (2021 - 2023)
 - Natural Language Processing (2026 spring, with Prof. Bhuwan Dhingra)
 - Web Data Mining (2023 fall, with Prof. Xiaojun Wan)
 - Introduction to Computing (Python, 2023 fall, with Prof. Xifeng Deng)
 - Data Structures and Algorithms (2022 spring, with Prof. Zhao Wang)
 - Introduction to Computing (C++, 2021 fall, with Prof. Yibo Lin)
- **Reviewer:** 2022 - Present
 - ACL-ARR (2023 - Now), ICLR (2024 - 2026), NeurIPS (2024 - 2026), ICML (2025 - 2026), AAI (2024 - 2025), COLM (2025), COLING (2024), JCST (2025)
- **Volunteer:** 2023 - Present
 - AAI'24, ACL'24, NLPCC'23 Shared Task 8 track chair

SKILLS

- Pytorch, Transformers, Distributed Training, Accelerated Inference
- Python, C++, HTML, JavaScript